



भारतीय सूचना प्रौद्योगिकी संस्थान गुवाहाटी
Indian Institute of Information Technology Guwahati

CS236 - ARTIFICIAL INTELLIGENCE LAB
ASSIGNMENT - 01

1. Compose a Python program that accomplishes the following:
 - i. Assign a value to a variable and display both the value and its data type.
 - ii. Assign values to two variables and execute these arithmetic operations: addition, subtraction, multiplication, and division.
 - iii. Assign a string to a variable and determine the string's length.
 - iv. Assign a string value to a variable, and then convert the variable's type to an integer.
2. Develop a Python program to interchange the values of two variables, both with and without utilizing a temporary variable.
3. Given the string `str = "I am a computer science student"`, perform the following:
 - i. Display the initial character of the string.
 - ii. Display the characters from the second to the fifth position.
 - iii. Display all characters starting from the third position.
4. Create a Python program that does the following:
 - i. Generate a list containing six elements and display the entire list.
 - ii. Display the elements of the list from the third position to the end of the list.
 - iii. Insert the element 40 into the middle of the list, and append the element 100 to the end of the list.
 - iv. Display the third, fourth, and fifth elements of the list.
5. Write a program to combine two lists into a single list and display it. Then, print the even numbers present in the merged list.
6. Develop a Python program that performs the following:
 - i. Create a tuple containing a mix of data types and display it.
 - ii. Create a tuple that repeats the word `programming` three times.
 - iii. Create two tuples and then concatenate them into a single tuple.
7. Develop a Python program that does the following:
 - i. Given the dictionary `person = {'first_name': 'John', 'last_name': 'Doe', 'age': 25, 'favorite_colors': ['blue', 'green'], 'active': True}`, display the dictionary.

- ii. Display the keys and values of the dictionary separately.
 - iii. Display the value associated with the second key.
- 8. Write a Python program that accepts a number as input and determines whether it is even or odd using an if-else statement.
- 9. Construct a program to display numbers from a list using loops.
- 10. Create a Python function that accepts a list and returns a new list containing only the unique elements of the original list.
- 11. Create a dictionary `student_grades` where keys are student names (strings) and values are their corresponding grades (integers). Write a program to:
 - i. Print the grade of a specific student.
 - ii. Add a new student and their grade to the dictionary.
 - iii. Update the grade of an existing student.
 - iv. Iterate through the dictionary and print the names of students who scored above a certain threshold (e.g., 90).
- 12. Given a dictionary `inventory` representing items and their quantities in a store, write a program that uses dictionary methods (`keys()`, `values()`, `items()`, `get()`) to:
 - i. Display all the items in the inventory.
 - ii. Display the quantities of each item.
 - iii. Check if a specific item is in stock and print its quantity (or a "Not found" message).
- 13. Write a function `calculate_rectangle_area` that takes `length` and `width` as parameters (with default values of 1) and returns the area of a rectangle. Demonstrate how to call the function:
 - i. With only positional arguments.
 - ii. With keyword arguments.
 - iii. Using the default values.
- 14. Write a program with a global variable and a function that defines a local variable with the same name. Demonstrate how the function accesses the local variable and the global variable can be accessed outside the function.
- 15. Write a function `process_student_data` that takes a list of dictionaries as input. Each dictionary represents a student and contains keys like `name`, `id`, and `grades` (where `grades` is a list of numbers). The function should:
 - i. Calculate the average grade for each student.
 - ii. Return a new dictionary where keys are student IDs and values are their average grades.